

Group29 Boom Boom Powell

Github repo :

<https://github.com/cse110-sp25-group29/warmup-exercise>

Youtube Link:

<https://youtu.be/BeC8TOHhnbQ>

SWOT Analysis

Background & Process

Our team is divided into four subgroups, Tutorial A/B and Intermediate A/B, based on each member's familiarity with HTML, CSS, and JavaScript. This structure is designed to help everyone engage with the code more effectively and learn through practical collaboration.

We have developed **two versions of the card game(Step1)** with different UI designs and animation styles, but sharing similar underlying logic. Also, we have done **Solitaire(Step1+2+3)** based on Step1.

Step1:

- **Version 1:** Cards flip on **hover** and flip back when the mouse leaves. At most one card can be face-up at a time. Clicking the Shuffle button triggers a shuffling animation.
- **Version 2:** Cards flip on **click**, and clicking again on the same card flips it back. Multiple cards can be face-up at once. Clicking the Shuffle button also triggers a shuffle animation.

Step1+2+3(Solitaire):

- Cards flip on **click**.
- Support **dragging** one or multiple cards from one column to another column.
- **Shuffle** all the cards to restart game.
- Follow the **basic rules of Solitaire**.

Strengths

- Minimalist and Clear Design: The definition of class is clean, with clearly defined pile, cloumn and card elements.
- Basic Interactivity: Card flipping (on hover or click, depending on version) and shuffle animations are already implemented, providing basic user engagement.
- Well-Structured Codebase: HTML, CSS, and JavaScript are properly separated in different folders in `static`, making the code easier to maintain and extend.
- Team Identity Reflected: The black and green theme reflects the unique branding of our team.
- Solitaire Mechanics Implemented: A functional Solitaire mode has been added, supporting core interactions like flipping, dragging, shuffling and other basic rules.

Weaknesses

- Limited Features(Solitaire):
 - Currently, the game lacks scoring, and win/lose conditions.
 - Lack of advanced UI and animations that could attract user's attention.

- Flip Mechanism Limitation(Step1):
 - Version 1: cards flip automatically on hover, which may reduce player control.
 - Version 2: cards flip on click, but the logic is still basic.
- Lack of Responsiveness: The game's compatibility with mobile devices or varying screen sizes is not yet fully optimized.

Opportunities

- Enhancing Solitaire Mode: Further developing Solitaire with full ruleset support, animations, and UI.
- Expand other functions: There's potential to incorporate memory matching, use login, competitive modes, or even multiplayer functionality.
- Adding Sound and Music: Implementing sound effects for actions like flipping, shuffling and scoring can enhance the immersive experience.
- Customizable Card Styles: Allowing users to personalize card appearances(both front and back) like upload their own images or choose from theme presets.

Threats

- Competitive to stand out: With more and more open-source card game already available, standing out without unique features can be challenging.
- Low Player Retention: A limited interactive experience might struggle to keep players engaged over time.
- Resource Limitations: Lack of access to design assets, audio, or backend(js) support may hinder future development and scalability.
- Complexity of Solitaire Implementation: Implementing full Solitaire rules and interactions (while keeping code maintainable) may pose a technical and time challenge.